

Integer Exponent Laws: Product and Quotient

Product of powers, quotient of powers, zero and negative exponents

The model. A power is a count of equal factors. Expanding the factors is what makes every law on this page obvious, so expand first and state the law second.

Power	Expanded form	Factors
2^3	$2 \cdot 2 \cdot 2$	three
2^4	$2 \cdot 2 \cdot 2 \cdot 2$	four
$2^3 \cdot 2^4$	$(2 \cdot 2 \cdot 2)(2 \cdot 2 \cdot 2 \cdot 2)$	three + four

Directions:

- Show the expansion or the exponent arithmetic — not just the answer.
- Leave every answer as a single power of the given base unless the problem asks for a number.
- Bases never change. Only the exponents move.

1. Expand both powers, then rewrite the product as one power of 3.

$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = \underline{\hspace{2cm}}$$

Answer: _____

2. Use the product law. Write the answer as a single power of x .

$$x^4 \cdot x^6$$

Answer: _____

- 3.** Write the quotient as a single power of 5. Expanding the top and the bottom and cancelling pairs is a good way to see why the law works.

$$\frac{5^7}{5^4}$$

Answer: _____

- 4.** Write as a single power of y .

$$\frac{y^9}{y^3}$$

Answer: _____

- 5.** Simplify $a^3 \cdot a^2$ first, then substitute $a = 2$ and give the numerical value.

Answer: _____

6. Simplify completely. The coefficients divide; the powers of x follow the quotient law.

$$\frac{12x^5}{4x^2}$$

Answer: _____

7. Two ways of reading the same expression:

$$\frac{6^4}{6^4}$$

(a) Cancel the identical factors on top and bottom. What number is left? (b) Apply the quotient law without cancelling: subtract the exponents and write the power that results. Your two answers must agree — write the value that 6^0 must therefore equal.

Answer: _____

8. Apply the quotient law to

$$\frac{4^2}{4^5}$$

even though the top exponent is smaller. Write the result as a power of 4 with a negative exponent, then write it again as a fraction with no negative exponent.

Answer: _____

9. Decide which expression is larger, or whether they are equal, and write $<$, $>$ or $=$ in the blank. Simplify each side to a single power of 3 before you compare — do not multiply the numbers out.

$$3^4 \cdot 3^2 \text{ _____ } \frac{3^7}{3^1}$$

Answer: _____

10. Jae simplified $2^5 \cdot 2^3$ by multiplying the exponents, writing $2^{5 \cdot 3}$, and answered 32768. Explain in one sentence which law Jae should have used, then give the correct value of $2^5 \cdot 2^3$.

Answer: _____